Problem I: Quantum propagator for harmonic oscillator (Ref. Feynman)

(1) Consider a harmonic oscillator $H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 x^2$. Obtain the classical path as

$$x(t) = x_0 \frac{\sin[\omega(t_f - t)]}{\sin[\omega(t_f - t_0)]} + x_f \frac{\sin[\omega(t - t_0)]}{\sin[\omega(t_f - t_0)]}$$

where x_0 and x_f are the initial and final position, respectively, and t_0 and t_f are the initial and final time, respectively.

(2) *The classical action is defined by $S = \int_{t_0}^{t_f} \mathcal{L}(v, x)dt = \int_{t_0}^{t_f} \frac{1}{2}mv^2 - \frac{1}{2}m\omega^2 x^2 dt$ where v is velocity. Use the classical trajectory to derive the classical action for the harmonic oscillator

$$S = \frac{1}{2}m\omega \left[\frac{(x_0^2 + x_f^2)\cos(\omega t_f - \omega t_0) - 2x_0 x_f}{\sin(\omega t_f - \omega t_0)}\right]$$

(3) Use the definition of the stability function in 1D system, $J(t) = \frac{dx(t)}{dp(0)}$ to obtain $J(t) = \frac{\sin(\omega t)}{m\omega}$ for the harmonic oscillator.

(4) In the lecture, the semiclassical (SC) propagator is given as $G(x_f, x_0, t) = \sqrt{\frac{1}{2\pi J(t) i}}\, e^{iS(t)}$

Use the expressions above to write the propagator for the harmonic oscillator.

(5) Explain why the semiclassical propagator is exact for the harmonic oscillator. Which of the following systems cannot be solved exactly with the semiclassical (SC) approximation?

i. Displaced harmonic oscillator
ii. Driven harmonic oscillator
iii. Non-linear oscillator in general
iv. Dissipative oscillator

Problem III: Quantum response theory

(Ref: McQuarrie, Kubo).

1) Define the quantum Liouville operator as $\mathcal{L}A = \frac{1}{\hbar}[H, A]$. Show that $\dot{A}(t) = i\mathcal{L}(t)A(t)$ and $\dot{\rho} = -i\mathcal{L}\rho$, where ρ is the density matrix and $A(t)$ is a Heisenberg operator.

2) Given $H = H_0 - AF(t)$, use perturbation theory to show

   $\langle A(t)\rangle = \int_{-\infty}^{t} K(t-t')F(t')dt'$ where $K(t-t') = \frac{i}{\hbar}\langle[A(t), A(t')]\rangle$.

3) Show that the classical limit of the quantum commutator is:

   $\frac{1}{i\hbar}[A, B] = \{A, B\}$, where $\{,\}$ is the Poisson bracket.

4) Show that the classical limit of $K(t)$ is $K(t) = -\beta\dot{C}(t)$.

Problem. IV: Linear and non-linear spectroscopy (Ref: Mukamel)

Spectroscopic measurements are expressed as polarization responses. We calculate the response function of a linear harmonic oscillator as an example.

1) The linear response function is defined as

$$\mathrm{R(t)} = \frac{\mathrm{i}}{\hbar}\langle 0|[\alpha(\mathrm{t}), \alpha(0)]|0\rangle = \frac{\mathrm{i}}{\hbar}[\langle 0|\alpha(\mathrm{t})\alpha(0)|0\rangle - \langle 0|\alpha(0)\alpha(\mathrm{t})|0\rangle],$$

where the transition dipole is assumed to be linear in coordinate $\alpha(\mathrm{t}) = \alpha_0 \mathrm{x(t)}$.

Show $\mathrm{R(t)} = \frac{\sin \omega \mathrm{t}}{\mathrm{m}\omega}\alpha_0^2$.

Useful expressions:

$\mathrm{x(t)} = \sqrt{\frac{\hbar}{2\mathrm{m}\omega}}\left(\mathrm{a}e^{-\mathrm{i}\omega \mathrm{t}} + \mathrm{a}^\dagger e^{\mathrm{i}\omega \mathrm{t}}\right), \langle 0|\mathrm{aa}^\dagger|0\rangle = 1, \langle 0|\mathrm{a}^\dagger \mathrm{a}|0\rangle = 0.$

2) Show that the same result can be obtained classically by replacing the quantum commutation with the Poisson bracket,

$$\frac{1}{\mathrm{i}\hbar}[\alpha(\mathrm{t}), \alpha(0)] \to \{\alpha(\mathrm{t}), \alpha(0)\} = \frac{\partial \alpha(\mathrm{t})}{\partial \mathrm{x}(0)}\frac{\partial \alpha(0)}{\partial \mathrm{p}(0)} - \frac{\partial \alpha(\mathrm{t})}{\partial \mathrm{p}(0)}\frac{\partial \alpha(0)}{\partial \mathrm{x}(0)}$$

Useful expression: $\mathrm{x(t)} = \mathrm{x}(0)\cos \omega \mathrm{t} + \frac{\mathrm{p}(0)}{\mathrm{m}\omega}\sin \omega \mathrm{t}$

3) The non-linear response function is defined as

$$\mathrm{R(t_1, t_2)} = \left(\frac{\mathrm{i}}{\hbar}\right)^2 \langle 0|\big[[\alpha(\mathrm{t}_1 + \mathrm{t}_2), \alpha(\mathrm{t}_1)], \alpha(0)\big]|0\rangle,$$

If $\alpha = \alpha_0 \mathrm{x}$, show $\mathrm{R(t_1, t_2)} = 0$ from the balance of a and $\mathrm{a}^\dagger$ operators.

4) To obtain non-vanishing non-linear response for the harmonic oscillator, we introduce a non-linear coordinate dependence, $\alpha = \alpha_0 \mathrm{x} + \alpha' \mathrm{x}^2 + \cdots$

Show that the leading order term in α' is proportional to $\alpha_0^2 \alpha'$.